IES ACADEMY

IES Academy

LESSON PLAN

For

GenAI for Engineers

Conducted by

IES Academy

Version 2.4

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.4	19 October 2026	Simple, self-contained labs: short steps with one prompt each, and every lab works directly on the files in its own lab folder (no extra folder, no copying, no dependence on earlier labs); files stay local with no OneDrive upload; new Lab 7 PowerPoint design skill (plain to professional deck); Failure_Mode column and failure-mode Pareto in Labs 5-6; slides on Copilot as the agentic AI interface and on SKILL.md anatomy; Word, PowerPoint and Outlook steps upload the local file with + > Upload images and files (paste as fallback); new slide compares local upload with OneDrive / reference. OneDrive remains only for the Excel Skills folder and Agent Builder knowledge.	IES Academy
2.3	18 October 2026	Lab 11 Agent Builder work-order agent; Lab 12 group mini capstone; Copilot interface, AI security and Agent Builder pages; corrected key concepts; built-in @chart-design demo; dashboard with KPI tiles and slicers; lean lab folders; step-by-step prompts page.	IES Academy
2.2	14 October 2026	Ten connected labs: account Personalisation, visible .Rules, create/apply Pareto and reporting Skills, custom interactive-dashboard Skill, filters and controlled refresh.	IES Academy
2.1	13 October 2026	Water-pipe service scenario; expanded coherent prompts, pure-text Copy buttons, linked PDFs and simple data/PDF labs.	IES Academy
1.0	29 July 2026	Previous cross-app scenario; retained in archive.	IES Academy
2.0	13 October 2026	Four maintenance topics; eight connected labs, custom Excel/PowerPoint skills, interactive dashboard and management presentation.	IES Academy

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TABLE OF CONTENTS

Course Information **4**
Learning Outcomes **4**
Learning Reinforcement **4**
Course Schedule **4**
Day 1 — Maintenance workflow, custom skills, interactive dashboard, management presentation, Copilot agent and group capstone 4
Lab Reference (aligned to course topics) **5**

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Course Information

Course Title	GenAI for Engineers
Training Provider	IES Academy
Duration	1 day · 7.5 instructional hours (8 scheduled hours excluding lunch, including tea breaks)
Daily Timing	9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)
Mode	Instructor-led, hands-on practical labs
Trainer	IES Academy Trainer

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Integrate Word, Excel, PowerPoint and Outlook Copilot in a coherent maintenance workflow with explicit source grounding and verified prompts.
- LO2: Distinguish account Personalisation, workbook .Rules and reusable Skills; configure, create, apply and verify Pareto and maintenance-reporting Skills using water-service evidence.
- LO3: Create and apply a custom interactive-dashboard Skill with verified measures, shared filter scope, editable charts and controlled refresh and rerun tests.
- LO4: Continue from the dashboard to a professional management presentation, with evidence-based headlines, source notes, rehearsal and a reviewed handoff draft.
- LO5: Build and test a grounded work-order agent with Agent Builder, then apply the full data-to-decision workflow to your own mock or sanitised data in a group capstone.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — Maintenance workflow, custom skills, interactive dashboard, management presentation, Copilot agent and group capstone

Time	Duration	Topic / Activity
09:30–09:40	10 min	Welcome, learning outcomes, app access and Northstar scenario
09:40–09:50	10 min	**Topic 1: concepts and demonstration**
09:50–10:20	30 min	Lab 1 (Excel to Word): Prepare Work Orders and Draft the Operations Brief
10:20–10:50	30 min	Lab 2 (Word to PowerPoint and Outlook): Create the Cross-App Maintenance Update
10:50–11:05	15 min	Tea break
11:05–11:30	25 min	**Topic 2: concepts and demonstration**

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Time	Duration	Topic / Activity
11:30–11:50	20 min	Lab 3 (Personalisation): Personalise Copilot in Excel for Consistent Service Reports
11:50–12:10	20 min	Lab 4 (Workbook .Rules): Create and Test Workbook .Rules for Water Service Reporting
12:10–12:40	30 min	Lab 5 (Custom Pareto Chart Skill): Find the Vital Few Failure Modes with a Pareto Skill
12:40–13:40	60 min	Lunch
13:40–14:10	30 min	Lab 6 (Reporting Skill): Build a Maintenance Report with Two Skills
14:10–14:40	30 min	Lab 7 (PowerPoint Design Skill): Create a PowerPoint Design Skill: From Plain to Professional
14:40–14:50	10 min	**Topic 3: concepts and demonstration**
14:50–15:25	35 min	Lab 8 (Custom Dashboard Skill): Build an Interactive Dashboard with a Custom Skill
15:25–15:40	15 min	Tea break
15:40–16:05	25 min	Lab 9 (Interactive Filters and Refresh): Test Dashboard Filters and a Data Refresh
16:05–16:10	5 min	**Topic 4: concepts and demonstration**
16:10–16:35	25 min	Lab 10 (Dashboard to presentation): Turn Dashboard Evidence into a Management Story
16:35–17:00	25 min	Lab 11 (Professional handoff): Refine and Present the Maintenance Decision Deck
17:00–17:05	5 min	**Topic 5: concepts and demonstration**
17:05–17:35	30 min	Lab 12 (Copilot Agent Builder): Build a Work-Order Agent with Agent Builder
17:35–18:20	45 min	Lab 13 (Mini Capstone): Mini Capstone: Group Dashboard and Decision Deck
18:20–18:30	10 min	LO1-LO5 recap, capstone reflections and questions

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 1: Fundamentals of Microsoft 365 Copilot	2 of 13 labs (15%)	Lab 1, Lab 2
Topic 2: Personalisation, .Rules and Skills for Excel	5 of 13 labs (38%)	Lab 3, Lab 4, Lab 5, Lab 6, Lab 7
Topic 3: Interactive Dashboards with Custom Skills	2 of 13 labs (15%)	Lab 8, Lab 9
Topic 4: Create a Professional Presentation with PowerPoint Copilot	2 of 13 labs (15%)	Lab 10, Lab 11
Topic 5: Copilot Agents and Mini Capstone	2 of 13 labs (15%)	Lab 12, Lab 13

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